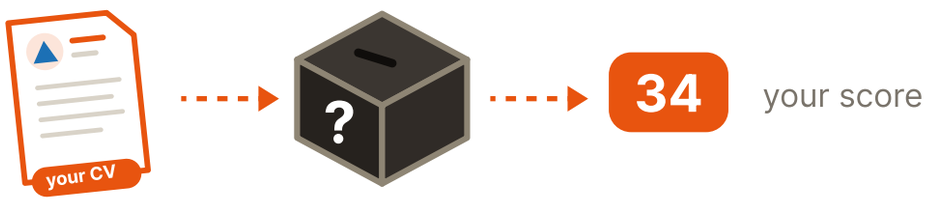
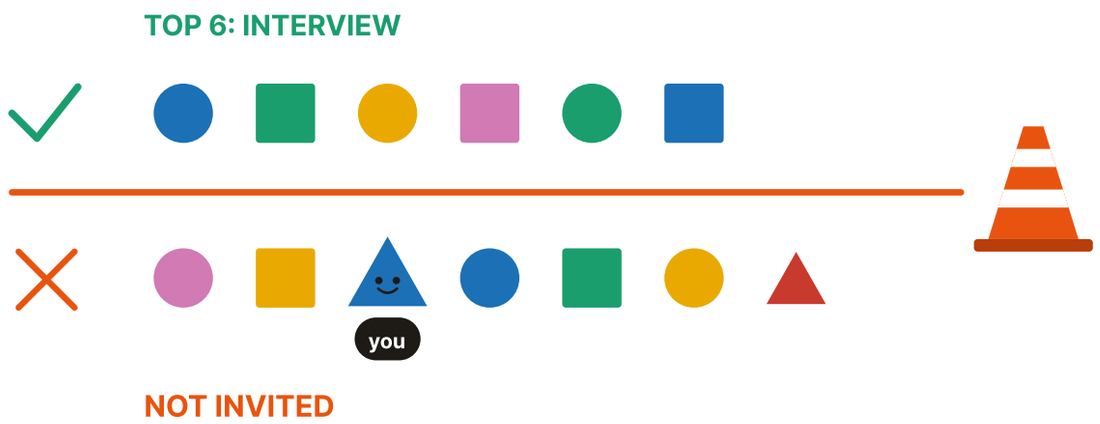
READ ME FIRST

Fairness and algorithms demo

The black box

Too many CVs, too little time. So a company bought an **algorithm: it gives every CV a score and then ranks every** candidate in a row. **The top 6 get an interview.**





Why is it called a ‘black box’?

You can see what goes in and what comes out, but what happens in between is too complex to follow. The box learned by itself, from **past hiring decisions**. Did people use to prefer circles (often without realising it)? Then the box probably learned that too. And it is **not perfect**: it is right only in about 7 out of 10 cases.

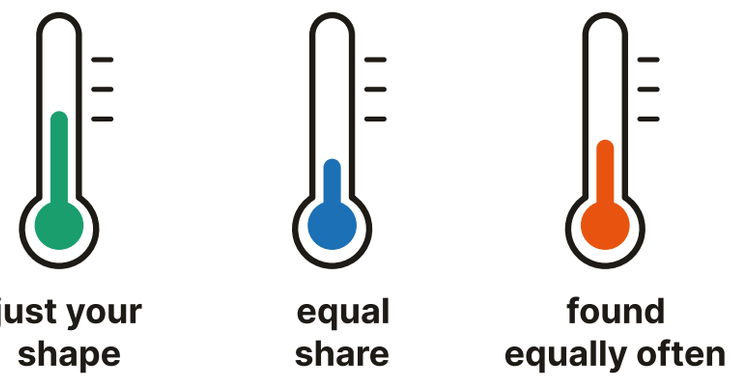
Removing sensitive attributes from the cv, also called **fairness through unawareness**, is not enough. Even if nobody tells the box your shape, it can infer it from the rest of your CV. Therefore we do measure the bias.



The thermometers

How do you know whether the algorithm is fair? There are tests for that: *fairness metrics*. They work like a **thermometer**: an abnormal reading means something is off.

There are **more than 50** of these tests. We show three. If a thermometer shows a deviation, the company can **adjust** the algorithm.
The problem: each thermometer looks at something different, and you can never get them all right at the same time. So someone has to choose which one counts most. And today, you get to do that!



1 · Just your shape

counterfactual fairness

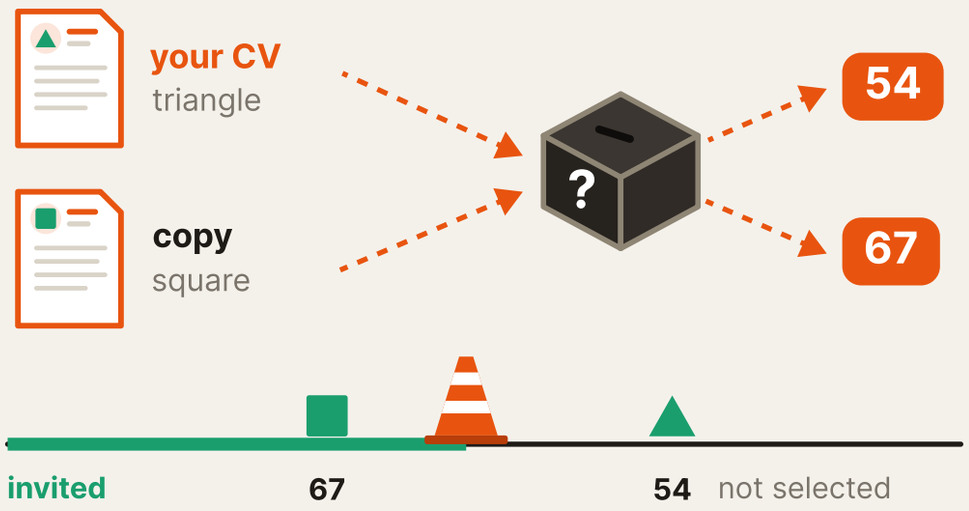
Would you get the same result if you only changed your shape or your size?

HOW TO MEASURE IT

1. Make a **copy** of your CV.
2. On the copy, change **only your shape or your size**, and whatever follows directly from that.
3. Let the algorithm score **both** CVs. If the scores differ, the box is paying attention to your shape after all.

HOW TO CORRECT IT

The box is adjusted until the original CV and the copy get **roughly the same score, for everyone.**



GOOD TO KNOW

> It looks at each person individually, not at groups. So one shape can still be invited less often. And what ‘changes along’ with your shape is often a discussion in itself.

HOW DO THE THERMOMETERS WORK?

2 · Equal share

statistical parity

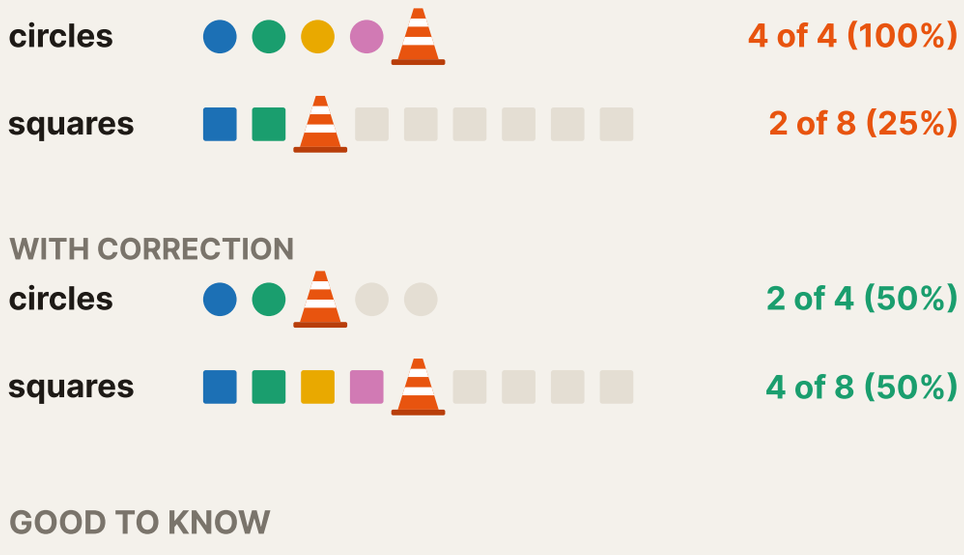
Does each shape get an equal share of interviews?

HOW TO MEASURE IT

This rule makes sure that **every shape has an equal chance at an interview.** Count per shape: how many applied, and how many were invited? If those shares are not roughly the same, the algorithm scores badly on this thermometer.

HOW TO CORRECT IT

Form **one row per shape** and invite the **same percentage** from each row. Each shape then gets its own cone, which can align with a different score.



GOOD TO KNOW

> This doesn’t just look at who is suitable: sometimes someone with a higher score stays home. Useful if you think past hiring or requirements are itself skewed.

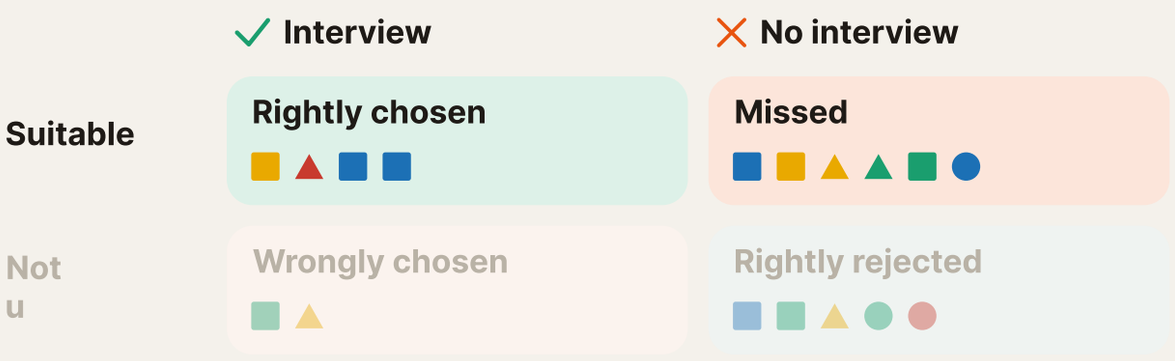
3 · Found equally often

equal opportunity

Does the algorithm find qualified people equally often for each shape?

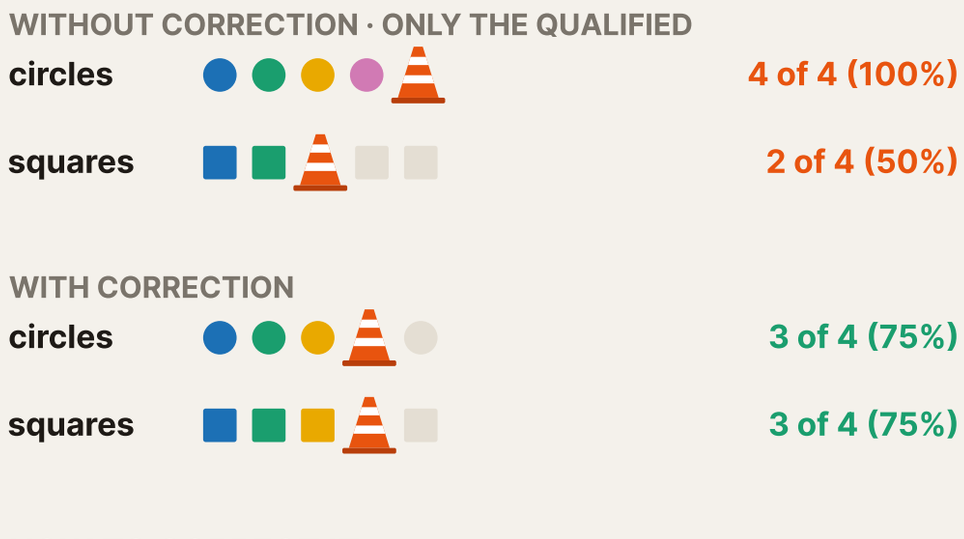
HOW TO MEASURE IT

Take applicants **from the past** whose real suitability you now know (e.g. an evaluation after 6 months). Look **only at the suitable row: who was rightly chosen, who was wrongly** missed? Count the proportions per shape.



HOW TO CORRECT IT

Move the cone **per shape** until each shape gets the same percentage of its qualified candidates.



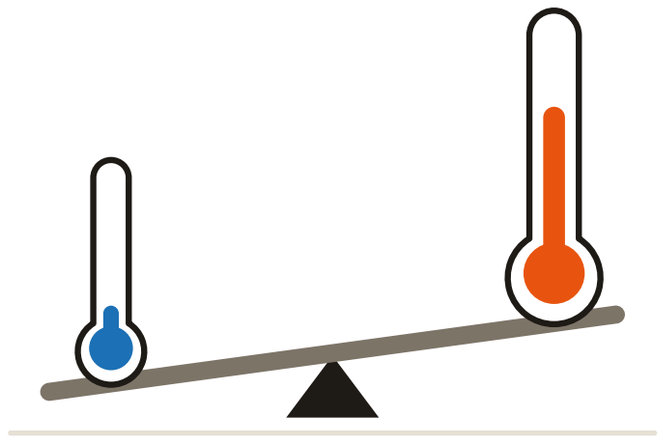
GOOD TO KNOW

> You need data from the past, which can be biased too. But above all: how do we decide what ‘suitable’ or ‘qualified’ means?

They can’t always all be right.

Suppose: in the past, **6 of the 8 circles** turned out to be suitable, but only **2 of the 8 squares. There are 6 places.**

- **‘Equal share’ wants 3 circles and 3 squares. So at least one square that was** not suitable gets through, while suitable circles stay home.
- **‘Found equally often’ wants the same share of the suitable candidates for both shapes: roughly 4 to 5 circles and 1 to 2 squares.**
- And **‘just your shape’**? The other two rules give each shape its own cone. So your outcome sometimes does change when your shape changes.



Which one gets priority is a choice about what you find fair. So in the app you always choose between two thermometers: **which one should always stay right, in your view?**

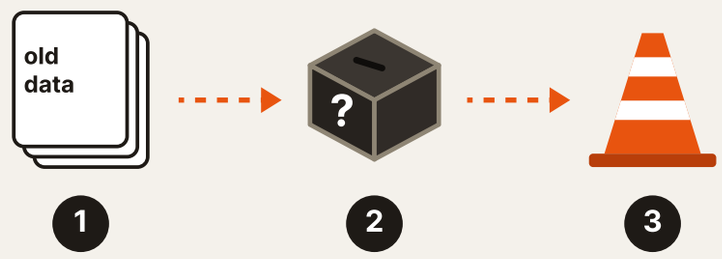
IN PRACTICE

From shapes to people

In reality, of course, it's not about red circles or yellow squares, but the same principles apply to characteristics you do know:

The shape & size you were given	**gender, age, skin color**
The colour you chose	**education, hobbies, circle of friends**
Hidden 'qualified'	**still working there after a year, a positive evaluation**

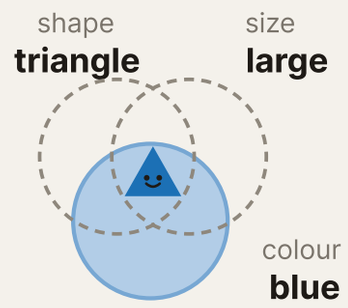
Where can you correct it?



1. **Before: fix the old data**, e.g. give examples of rarely hired shapes more weight.
2. **During: the algorithm learns with an extra rule**: 'keep this thermometer right'.
3. **After: the algorithm stays while the cone moves**, e.g. to a different score per shape.

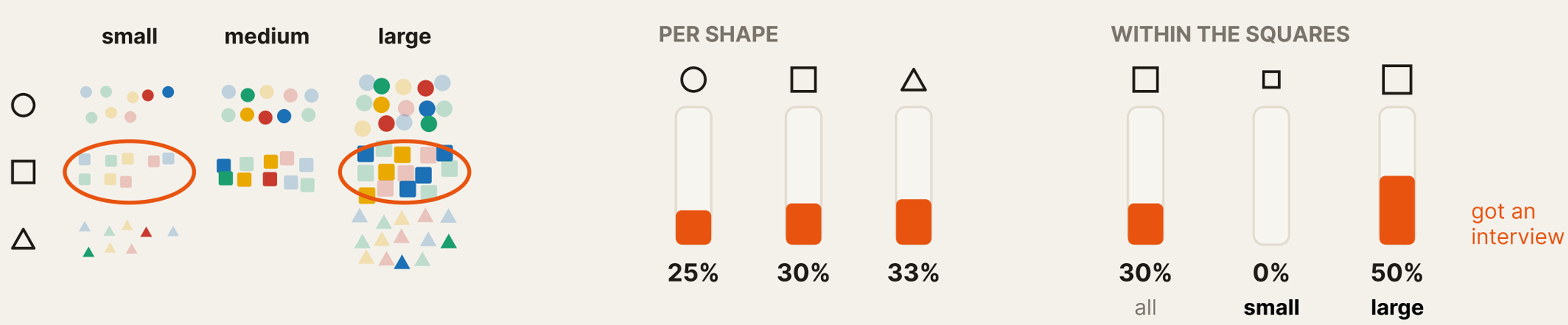
THE CORNER NOBODY LOOKED AT

You are more than one group.



The thermometers looked at only one thing, such as your shape. But you are also large, a triangle, and you chose blue. **You are in all those groups at once.**

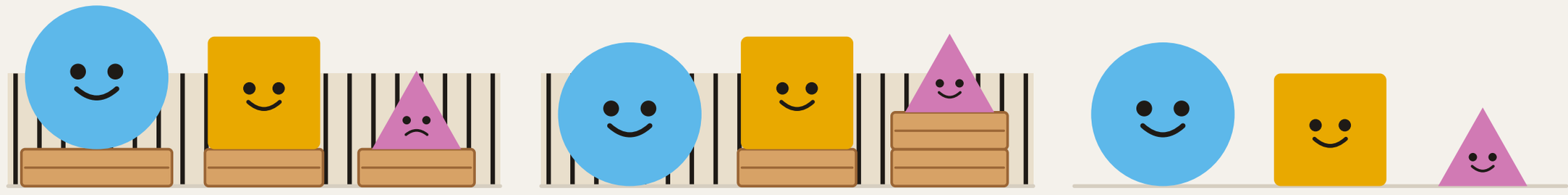
Take 100 applicants and group them by shape and size, with 'equal share' as the rule. **Per shape it looks fair: each shape got interviews in proportion. But within the squares,** 10 of the 20 large ones got an interview, compared to 0 of the 15 small ones.



This is called **intersectionality**: you belong to several groups at once, and where they cross, a metric can miss you.

The rule did more than measure.

A thermometer only measures. But 'equal share' and 'found equally often' also changed who got an interview: if a shape scored low, it sometimes got a nudge upwards.



Equality

Everyone gets the same box, just like the unadjusted algorithm uses one rule for everyone. Sounds fair, but the smallest still can't see over the fence.

Equity

Everyone gets the support they need. Now everyone can see over the fence. That's how 'equal share' and 'found equally often' worked.

Justice

The fence comes down. Then nobody needs a box any more. But how do you get that fence down?

JUST SORT FAIRLY, OR ALSO REPAIR?

Those low scores come from somewhere.

The algorithm learned from old hiring decisions. Some shapes got fewer chances for a long time, and that now shows in the numbers. Sorting fairly redistributes today's interview places. It does nothing about what grew skewed.

> The boxes help you see over the fence. But **the fence is still there.**

Another question: repair.

For some researchers, fair sorting is not enough. Their question is not 'who gets invited today?' but 'who was disadvantaged before, and how do we make it right?' For example: if two candidates are equally qualified, the one from the disadvantaged group gets the spot.

> This is called **justice**: solving the cause, so that in time nobody needs a box any more.

Does this really happen?

Yes. Algorithms help decide on job applications, loans, insurance and benefits. A well-known example: Amazon stopped using a CV algorithm because it scored women lower. The **European AI Act** classifies **AI for hiring as 'high risk': such systems must, among other things, be checked more closely for bias.**
But which thermometer (= 'metric') counts is currently mostly chosen by the developers of algorithms. Our goal is for the people affected by the outcomes to have a say as well. So feedback on clarity is always welcome!

THIS DEMO WAS MADE BY

SMIT
VUB · imec

AUGMENT
KU Leuven · imec

Questions? Feedback?
→ an.jacobs@vub.be, anna.verheyden@kuleuven.be
Want to do this workshop as a group?
→ an.jacobs@vub.be

Organising it yourself

To organise it yourself, you need:
- **Wi-Fi**
- Enough **devices** to participate (phones, laptops or tablets).
  You'll find the app at https://fairness-demo-fti.vercel.app/
- OPTIONAL: **tables** with the physical set-up, **blocks** as 'personas', **cards**, **stamps**

Useful links

The web app:
https://fairness-demo-fti.vercel.app/

The buttons for voting on what is fair, in order:
- https://fairness-demo-fti.vercel.app/tablet/bare_box
- https://fairness-demo-fti.vercel.app/tablet/counterfactual
- https://fairness-demo-fti.vercel.app/tablet/parity
- https://fairness-demo-fti.vercel.app/tablet/opportunity

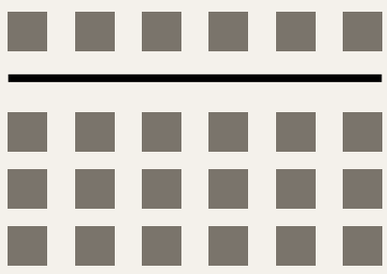
(password 'fti-demo')
If you want to do it without voting tablets, put /solo behind the url.

Persona

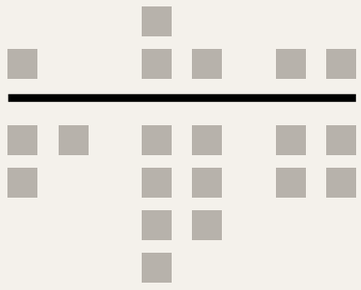
Squares	small: 5 large: 5
Circles	small: 2 large: 2
Triangles	small: 3 large: 3

Table set-up

The algorithm, thermometer 1

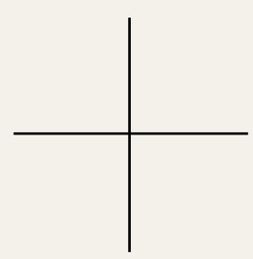


Thermometer 2



The number of people per shape differs, so e.g., squares have more places

Thermometer 3



The ranking follows the same layout as thermometer 2, but now only the top two cells of this grid count.

Cards

After each selection, a participant can stamp whether or not they were selected for an interview.
At the end, participants hang this card on a board divided in two: on the left if they choose fair sorting without repair, on the right if they think repair does matter. You can add extra questions on the back if you like.

